Perpendicular lines

1 Describe what it means for two lines to be perpendicular.

2 If two lines are perpendicular, state the product of their gradients.

3 State whether the following pairs of lines are perpendicular:

a $y = x + 1$ and $y = x - 1$

b $y = 3x - 7$ and $y = -3x + 6$

c $y = -3x + 6$ and $y = \frac{x}{3} + 7$

d $y = \frac{2x}{3} + 4$ and $y = -1 - \frac{3x}{2}$

e $y = -\frac{x}{3} - 9$ and $y = -(3x - 27)$

f $y = 3x - 5$ and $y = -\frac{1}{3}x + 6$

g $y = -\frac{2x}{3} + 1$ and $y = -1 + \frac{3x}{2}$

h $y = \frac{3x}{5} + 6$ and $y = \frac{-5x - 30}{-3}$

4 Given that the following pairs of lines are perpendicular, calculate the value of m:

a $5x + 4y + 8 = 0$
$mx + 9y - 8 = 0$

b $-6x + 6y - 8 = 0$
$mx + 10y + 8 = 0$

5 Find the gradient of the line perpendicular to the following lines:

a Line with gradient 6

b $y = 6x + 3$

c $y = \frac{x}{8} + 3$

d $y = 8 - x$

6 Consider the lines L_1: $8x = -7y + 3$ and L_2: $7y - 8x - 7 = 0$.

a Find the gradient of line L_1.

b Find the gradient of line L_2.

c Are the two lines perpendicular?

7 Find the equation of a line described by the following information:

a Perpendicular to the x-axis and passes through $(-8, -1)$.

b Perpendicular to the y-axis and passes through $(-8, -8)$.

c Perpendicular to $y = -\frac{x}{2} + 5$, and goes through the point $(0, 6)$.

d Perpendicular to $y = 6x + 10$, and has the same y-intercept.

e Perpendicular to the line that passes through $(1, 1)$ and $(3, 13)$, and has a y-intercept of $(0, -4)$.

8 The line L_1 is perpendicular to $y = 5x - 4$ and cuts the y-axis at 1.

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a Find the gradient of line L_1.

b Find the equation of line L_1.

9 The line L_1 is perpendicular to $y = 10x + 8$ and passes through the point of intersection of the lines $y = x + 5$ and $8x - 10y + 60 = 0$.

a Find the gradient of line L_1.

b Find the point of intersection of $y = x + 5$ and $8x - 10y + 60 = 0$.

c Find the equation of the perpendicular line L_1.

Applications

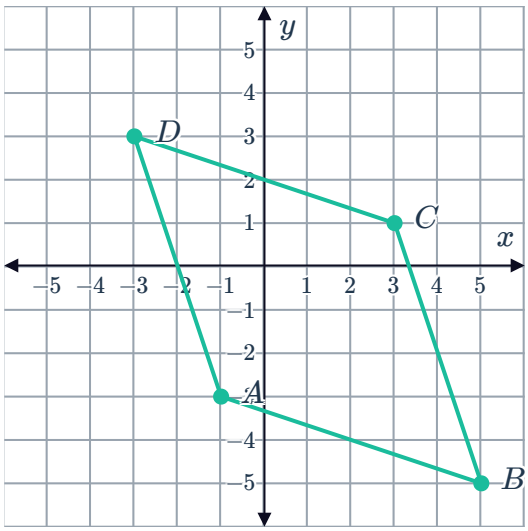
10 Consider the rhombus $ABCD$ on the number plane:

a Find the gradients of its diagonals.

i Gradient of AC

ii Gradient of BD

b Are the diagonals of the rhombus perpendicular? Explain your answer.

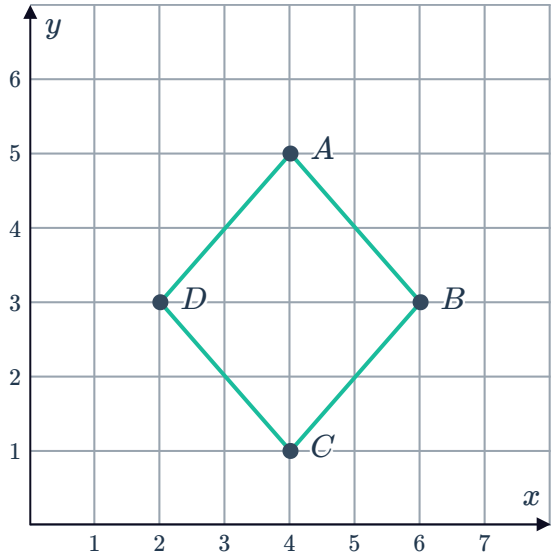


11 The four vertices of quadrilateral $ABCD$ have been plotted on the number plane:

a Find the gradient of the following sides:

i AB **ii** BC **iii** DC **iv** AD

b Given that side $AB = BC$, determine the type of quadrilateral described by the four points.



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